Journal of the Operational Research Society (2001) 52, 708–717 0160-5682/01 $15.00

www.palgrave-journals.com/jors

NP-hard cases in scheduling deteriorating jobs on dedicated machines

A Kononov[1] and S Gawiejnowicz[2]*

[1]*Sobolev Institute of Mathematics, Novosibirsk, Russia and* [2]*Adam Mickiewicz University, Poznań, Poland*

In this paper problems of time-dependent scheduling on dedicated machines are considered. The processing time of each job is described by a function which is dependent on the starting time of the job. The objective is to minimise the maximum completion time (makespan). We prove that under linear deterioration the two-machine flow shop problem is strongly NP-hard and the two-machine open shop problem is ordinarily NP-hard. We show that for the three-machine flow shop and simple linear deterioration there does not exist a polynomial-time approximation algorithm with the worst case ratio bounded by a constant, unless $P = NP$. We also prove that the three-machine open shop problem with simple linear deterioration is ordinarily NP-hard, even if the jobs have got equal deterioration rates on the third machine.

Keywords: deteriorating jobs; flow shop; open shop; makespan; NP-hardness

Introduction

In the classical formulation of the scheduling theory it is assumed that the processing times of jobs, as well as other parameters of a set of jobs, are *constant* values known in advance (see, eg, Conway *et al*[1]). This assumption, however, is inadequate for the modelling of many modern industrial processes where very often a job, executed in the same or almost the same conditions, has a variable processing time. It takes place, for example, when the machine used is losing efficiency during work, causing the job processing times to increase. In the literature the phenomenon is called the *deterioration* of job processing time.

There are different approaches to the problem of deterioration. In one approach the function that describes the processing time of a job depends on some *expression* depending in turn on the starting time of the job. This approach has been introduced by Sriskandarajah and Goyal,[2] and developed by Sriskandarajah and Wagneur[3–5] and Finke and Jiang.[6]

Using this approach, Sriskandarajah and Wagneur[3] have formulated a two-machine flow shop problem. The processing times of operations executed on the first machine are constant, while the processing times of operations performed on the second machine are variable and equal to the sum of a constant part and a function of the time difference between the completion time of the operation executed on the first machine and the starting time of the operation executed on the second machine. The function that depends on the difference can be a simple linear function[2] or an arbitrary nondecreasing function.[3] It has been proved[2,3] that the problem of minimising the makespan for a two-machine flow shop with processing times defined as above is NP-hard in the strong sense both for simple linear and arbitrary nondecreasing functions describing variable processing times. These results have been generalised[4] for a m-machine flow shop: as in the first case, the processing times of operations on the first machine are constant, while variable processing times of operations executed on the second and following machines are described by arbitrary nondecreasing functions of the differences defined as previously. For this problem results have been obtained[5] also for other than $C_{\max}$ (eg $L_{\max}$, $T_{\max}$) criteria.

Finke and Jiang[6] have formulated a problem which can be treated as the reverse to the above one. The assumptions are similar to those in the previous case with one exception: now the processing times of operations on the *last* machine are constant. It has been proved[6] that if the schedule length is the performance criterion, then a schedule is optimal for this problem if and only if the corresponding schedule is optimal for the previous problem. (By a *corresponding* schedule to the given one we understand here the schedule obtained by reading the given schedule from end to beginning, starting from the last machine and finishing with the first one.) If we deal with any regular performance criterion, then there exists an optimal permutation schedule of this problem for which all intervals of idle time are essential. (An interval of machine idle time is said to be *essential* if it is imposed by a sequence of operations which require non-overlapping processing times.)

*Correspondence: *Dr S Gawiejnowicz, Faculty of Mathematics and Computer Science, Adam Mickiewicz University, ul. Matejki 48/49, 60-769 Poznań, Poland.*
E-mail: stgawiej@amu.edu.pl

The models that use the above two approaches have many industrial applications, especially in environments where the jobs cannot be processed on machines until the moment when they are heated up to a certain temperature.[2] In this case the processing times depend on temperature. Other applications are, eg, in steel production, plastic molding, silverware production, chemical or pharmaceutical industries, in machines maintenance and servicing[3–6] and in other environments in which the processing time of a job depends on waiting time or on the time spent prior to processing.

In this paper we focus our attention on the problems in which the processing time of a job is described by an increasing function that depends only on the starting time of the job. This stream of scheduling theory has been initiated by Melnikov and Shafransky.[7] As we are interested in the functions that are either linear or proportional to a linear function, we review here only the results that are related to these forms of deterioration. Other kinds of time-dependent scheduling problems are reviewed by Alidaee and Womer.[8] More general scheduling problems in which not only processing times but also other parameters of a set of jobs are described by continuous dependencies are discussed by Gawiejnowicz.[9]

Most of the research that has been done in time-dependent scheduling deals with single machine problems, studied under the assumption that processing time of each job is described by a simple linear or a linear increasing function (see Browne and Yechiali,[10] Gawiejnowicz and Pankowska,[11] Kononov,[12,13] Mosheiov[14,15]) or a non-linear function (see Alidaee,[16] Gupta and Gupta,[17] Kononov,[18,19] Kubiak and van de Velde,[20] Kunnathur and Gupta,[21] Sundararaghavan and Kunnathur[22]) of the starting time of the job. Cheng and Ding[23] have considered single processor scheduling problems with non-zero ready times and deadlines. In all these cited papers the optimality criterion was the makespan but there are also known results concerning other criteria such as $L_{\max}$ or $\sum w_j C_j$ (see Kononov,[13] Mosheiov[15]). The case of parallel identical machines is discussed in a few papers (see Chen,[24] Kononov,[12] Mosheiov[25]). Results concerning time-dependent jobs and dedicated machines are not known, to the best of our knowledge, with two exceptions. In one paper, Melnikov and Shafransky[7] consider the two-machine flow shop problem with processing times described by the same function for all jobs subject to $C_{\max}$. In the second paper, Kononov[12] proves that for the case of simple linear processing times the three-machine flow shop problem subject to $C_{\max}$ is strongly NP-hard, while the two-machine flow shop problem and the two-machine open shop problem subject to $L_{\max}$ as well as the three-machine open shop problem subject to $C_{\max}$ are ordinarily NP-hard. He also shows that the two-machine flow shop problem and the two-machine open shop problem subject to $C_{\max}$ are polynomially solvable.

Time-dependent scheduling of this kind is applied, for example, in the area of scheduling maintenance or cleaning assignments where any delay implies an additional effort to finish the job.[15] This approach can be also applied in searching for an object under worsening weather conditions or growing darkness, in performance of medical procedures,[15] in scheduling chemical or metallurgical processes,[17] in crime scene response scheduling,[21] in scheduling emergency medical response teams, fire fighting,[22] etc.

Formulation of the problem

We now formulate the problem of scheduling time-dependent jobs on dedicated machines subject to the minimisation of the maximum completion time (makespan). We begin with the flow shop scheduling problem. A flow shop consists of a set of different machines that perform operations of jobs. We are given a set of n jobs $J_1, J_2, \ldots, J_n$ that have to be processed on the machines $M_1, M_2, \ldots, M_m$ of flow shop. Job J_j is composed of m operations $O_{1j}, \ldots, O_{mj}$, $j = 1, 2, \ldots, n$, with processing times p_{ij} that are functions of starting times S_{ij} of these operations, $p_{ij} = f_{ij}(S_{ij})$. Throughout the paper we will consider deteriorating processing times of three forms: simple linear, $p_{ij} = b_{ij}S_{ij}$, linear, $p_{ij} = a_{ij} + b_{ij}S_{ij}$, and proportional, $p_{ij} = b_{ij}(a + bS_{ij})$ where $a_{ij} \geqslant 0$, $b_{ij} \geqslant 0$, $a \geqslant 0$ and $b \geqslant 0$. For given i, j the coefficients a_{ij} (or a) and b_{ij} (or b) will be called *basic processing time* and *deterioration rate*, respectively.

All jobs have the same processing order through the machines of flow shop (ie the operation O_{ij} has to be processed on machine M_i) and are ready for processing starting from some time $t_0 \geqslant 0$. We also assume that there are no precedence constraints among operations of different jobs (all jobs are independent), operations cannot be interrupted (no preemption is allowable), each machine can handle only one job at a time, and each job can be performed on one machine only. A schedule for the problem is a set of starting times of particular jobs and of indices of machines that perform the jobs. The problem then is to find the schedule that minimises the makespan $C_{\max}$, ie the maximum of the completion time of all operations, $C_{\max} = \max_{ij}\{C_{ij}\}$, $1 \leqslant i \leqslant m$, $1 \leqslant j \leqslant n$. The formulation of an open shop scheduling problem is the same as for the flow shop problem except that the order of processing the operations within one job may be arbitrary.

The notation

Through the paper, $J \supseteq \{1, 2, \ldots, n\}$ will denote the set of job indices, $M = \{1, 2, \ldots, m\}$ the set of machine indices. C_{ij} will denote the completion time of the jth operation on machine M_i, while C_j the completion time of the jth job, $i \in M$, $j \in J$. The completion time of the last job executed on machine M_i will be denoted by $C(M_i)$, $i \in M$. $|X|$ will denote the number of elements of a set X. If some parameters depend on a given schedule we will denote it more clearly. For instance, $S_{1k}(s)$ denotes the starting time of the operation of job J_k executed in schedule s on machine M_1. Other notations will be explained at the place of their first appearence.

Describing scheduling problems we follow the well known notation $\alpha \mid \beta \mid \gamma$ (for details see, eg, Blazewicz *et al*,[26] section 3.4) with the following extension: in the field β the symbol $p_{ij} = f_{ij}(t)$ means that the processing time p_{ij} of the jth operation on the ith machine is described by a function $f_{ij}(t)$ of its starting time. (We will write t instead of S_{ij} to simplify notation, taking into account the fact that the starting time is a variable on which functions describing processing times depend.)

The organisation of the paper is as follows. The next section is devoted to the polynomially solvable cases and the strongly NP-hard cases of the two- and three-machine flow shop problem with simple linear or linear processing times as well as with processing times proportional to a linear function. We will show, in particular, that for the three-machine flow shop problem there does not exist a polynomial-time approximation algorithm with the worst case ratio bounded by a constant. The subsequent section covers similar results concerning the open shop problem. The final section includes conclusions and suggestions of the future research.

Flow shop problem

In this section we will establish the time complexity of two- and three-machine flow shop problems with simple linear processing times and with processing times proportional to a linear function.

We start with a lemma concerning the value of the completion time of a set of jobs with proportional processing times, scheduled on a single machine (thus for the case of this lemma we simplify the notation and write p_j, b_j instead of p_{ij}, b_{ij}).

Lemma 1 (Kononov[19]) *Let n jobs with processing times in the form of $p_j = b_j(a + bt)$ be executed on a single machine without idle times, starting from some time $t_0 \geqslant 0$. Then the completion time C_n of the last job in a schedule is equal to*

$$C_n \begin{cases} t_0 + a\sum_{j=1}^{n} b_j & \text{for } b = 0 \\ \left(t_0 + \dfrac{a}{b}\right)\prod_{j=1}^{n}(b_j b + 1) - \dfrac{a}{b} & \text{for } b \neq 0 \end{cases} \qquad (1)$$

and does not depend on the sequence of executed jobs.

Note that the result obtained by Mosheiov,[15] to the effect that the makespan for a single processor and simple linear processing times is independent of the sequence of jobs, is a special case of Lemma 1 (set $a = 1$ and $b = 0$).

It is a well known fact that for the case of fixed processing times of jobs, $C_{\max}$ criterion and at most three machines there exists an optimal flow shop schedule that is a permutation schedule, ie one in which all machines perform the jobs in the same order (see, eg, Blazewicz *et al*,[26] section 7.1.1 for details). For deteriorating processing times of jobs, however, this does not hold even for two machines. Consider the following example. We are given two jobs: job J_1 with processing times $p_{11} = 1 + 7t$, $p_{21} = 2$ and job J_2 with processing times $p_{12} = 1$, $p_{22} = 2t$. Both jobs are available for processing starting from time $t_0 = 0$. The optimal sequence for machine M_1 is (J_1, J_2), and for machine M_2 the sequence is reversed. Hence the optimal schedule is not a permutation schedule. We will now show that for the case of proportional processing times, $p_{ij} = b_{ij}(a + bt)$, there exist optimal schedules that are permutation schedules.

Lemma 2 *There exists an optimal schedule for the problem $Fm \mid p_{ij} = b_{ij}(a + bt) \mid C_{\max}$ in which machines M_1 and M_2 perform jobs in the same order.*

Proof: Assume that in schedule s jobs executed on machine M_1 are processed in the order $\pi_1 = (i_1, \ldots, i_n)$ and jobs executed on machine M_2 are processed in the order $\pi_2 = (j_1 = i_1, j_2 = i_2, \ldots, j_p = i_p, j_{p+1} = i_{r+1}, \ldots, j_{p+q} = i_r, \ldots, j_n)$, $q > 1, r > p \geqslant 0$.

By Lemma 1, the following equations are satisfied for schedule s:

$$S_{2,i_{r+1}}(s) = \max\left\{\left(\frac{a}{b} + t_0\right)\prod_{k=1}^{r+1}(b_{1,i_k}b + 1) - \frac{a}{b}, C_{2,j_p}(s)\right\}$$

$$\begin{aligned} S_{2,i_r}(s) &= \max\left\{\left(\frac{a}{b} + t_0\right)\prod_{k=1}^{r}(b_{1,i_k}b + 1) - \frac{a}{b}, C_{2,j_{p+q-1}}(s)\right\} \\ &= C_{2,j_{p+q-1}}(s) \\ &\geqslant C_{2,i_{r+1}}(s) \geqslant C_{1,i_{r+1}}(s) = \left(\frac{a}{b} + t_0\right)\prod_{k=1}^{r+1}(b_{1,i_k}b + 1) - \frac{a}{b} \end{aligned}$$

Consider schedule s', differing from schedule s only in that machine M_1 performs jobs in the order $\pi_3 = (i_1, \ldots, i_{r-1}, i_{r+1}, i_r, i_{r+2}, \ldots, i_n)$.

For schedule s', using Lemma 1, we obtain the following

$$S_{2,i_{r+1}}(s') = \max\left\{\left(\frac{a}{b} + t_0\right)(b_{1,i_{r+1}}b + 1)\prod_{k=1}^{r-1}(b_{1,i_k}b + 1) - \frac{a}{b}, C_{2,j_p}(s')\right\}$$

$$S_{2,i_r}(s') = \max\left\{\left(\frac{a}{b} + t_0\right)\prod_{k=1}^{r+1}(b_{1,i_k}b + 1) - \frac{a}{b}, C_{2,j_{p+q-1}}(s')\right\}$$

Since $C_{2,j_p}(s) = C_{2,j_p}(s')$, we have $S_{2,i_{r+1}}(s) \geqslant S_{2,i_{r+1}}(s')$. From this it follows that $C_{2,j_{p+q-1}}(s) \geqslant C_{2,j_{p+q-1}}(s')$ and hence $S_{2,i_r}(s) \geqslant S_{2,i_r}(s')$.

Repeating the above considerations no more than $O(n^2)$ times, we obtain a schedule $\bar{s}$ in which machines M_1 and M_2 perform jobs in the same order, and such that inequality $C_{2i}(s) \geqslant C_{2i}(\bar{s})$ holds for all $J_i \in J$. From this it follows that $C_{\max}(s) \geqslant C_{\max}(\bar{s})$. ■

Using the same argument as previously we obtain the following.

Lemma 3 *There exists an optimal schedule for the problem $Fm \mid p_{ij} = b_{ij}(a + bt)|C_{\max}$ in which machines M_{m-1} and M_m perform jobs in the same order.*

The next result follows immediately from Lemmas 2 and 3.

Lemma 4 *Let m be the number of machines, $m = 2, 3$. Then there exists an optimal schedule for the problem $Fm|p_{ij} = b_{ij}(a + bt)|C_{\max}$ in which jobs are executed on all machines in the same order.*

Now we will give the formulation of a strongly NP-complete 3-PARTITION problem (Garey and Johnson,[27] p 224).

3-PARTITION. Given a set $A = \{1, 2, \ldots, 3n\}$, an integer E, and positive integers $e_i, i \in A$, such that $\frac{E}{4} < e_i < \frac{E}{2}$ and $\sum_{i \in A} e_i = nE$, does there exist a partition of set A into n disjoint subsets A_j such that $\sum_{i \in A_j} e_i = E$, $j = 1, 2, \ldots, n$?

Now we may prove that the two-machine flow shop problem with linear processing times, $p_{ij} = a_{ij} + b_{ij}t$, is computationally intractable.

Theorem 1 *The problem $F2|p_{ij} = a_{ij} + b_{ij}t|C_{\max}$ is NP-hard in the strong sense.*

Proof: We will show that the 3-PARTITION problem is reducible to the problem $F2|p_{ij} = a_{ij} + b_{ij}t|C_{\max}$. Let $t_0 = 0$, $|J| = 4n$ and let jobs have processing times p_{ij} defined as follows:

$$\begin{array}{lll} p_{11} = 0 & p_{21} = 1 + nt & \\ p_{1j} = E + 1 & p_{2j} = \dfrac{1}{(j-1)(E+1)}t & 2 \leqslant j \leqslant n \\ p_{1j} = 0 & p_{2j} = e_j & n + 1 \leqslant j \leqslant 4n \end{array}$$

The jobs $J_{n+1}, \ldots, J_{4n}$ will be called *basic* jobs, while other jobs are *additional* ones. For simplicity we will call also the jobs with fixed processing times *F-jobs*. The threshold value is $y = nE + n$. We will prove that the 3-PARTITION problem has a solution if and only if the makespan for the problem $F2|p_{ij} = a_{ij} + b_{ij}t|C_{\max}$ is not greater than $nE + n$.

($\Rightarrow$) Assume that the 3-PARTITION problem has a solution. Then there exists a partition of the set $A = \{e_1, \ldots, e_{3n}\}$ into n three-element subsets $A_1, \ldots, A_n$ such that $\sum_{i \in A_j} e_i = E$, for $j = 1, \ldots, n$.

Consider the schedule presented in Figure 1. On machine M_1 we schedule only F-jobs. We obtain $C_{11} = 0$, $C_{1j} = S_{1j} + p_{1j} = (j-2)(E+1) + E + 1 = (j-1)(E+1)$ for $j = 2, \ldots, n$. Thus $C_{1n} = (n-1)(E+1)$. In scheduling jobs on machine M_2 we can use the same order, which implies that $C_{21} = 1$, $C_{2j} = C_{1j} + p_{2j} = (j-1)(E+1)+1$. Because $C_{2j} < C_{1,j+1}$ there are $n-1$ time intervals on machine M_2, each equal to E, between C_{2j} and $C_{1,j+1}$, $j = 1, \ldots, n-1$. In these intervals we can schedule $n-1$ triplets of jobs that are defined by elements from sets $A_1, A_2, \ldots, A_{n-1}$. The last such triplet, A_n, is scheduled after $C_{2,n} = (n-1)(E+1) + 1$. The makespan for the final schedule is then equal to $(n-1)(E+1)$ on machine M_1, and $(n-1)(E+1) + 1 + E = nE + n$ on machine M_2. Thus the final makespan is equal to $nE + n$.

($\Leftarrow$) Assume now that there exists a schedule s such that the makespan for the problem $F2|p_{ij} = a_{ij} + b_{ij}t|C_{\max}$ is not greater than $nE + n$. We will show that then the 3-PARTITION problem has a solution.

Note that because on machine M_1 we process only F-jobs and all the jobs have the same processing time equal to $E + 1$, the order of the jobs is immaterial. Moreover, the completion time $C_{1j} = (j-1)(E+1)$, $j = 2, \ldots, n$.

We will prove first that in schedule s the second operations of jobs J_{2j}, $j = 1, 2, \ldots, n$, will start exactly at time $(j-1)(E+1)$ and their lengths are equal to 1. Because $\sum_{j=n+1}^{4n} p_{2j} = nE$, the total processing time of second operations of jobs J_{2j}, $j = 1, 2, \ldots, n$, is not greater than n. Let permutation π define the order of performance of additional jobs on machine M_2. It is easy to see that job J_1 should be executed as the first in π; otherwise $p_{21} \geqslant 1 + n(E+1)$. Because the length of the first operation of each additional job, excluding O_{11}, is equal to $E + 1$, the kth job in permutation π will start not earlier than at $(k-1)(E+1)$. Without loss of generality we can

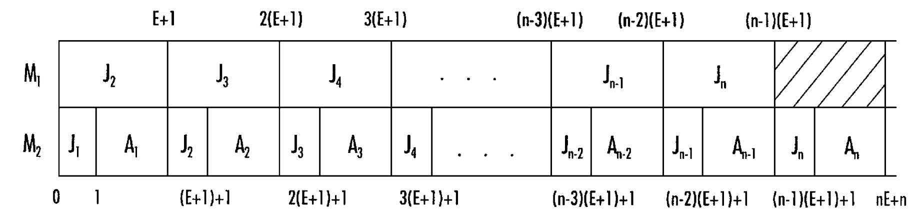

Figure 1 Example schedule for the problem $F2|p_{ij} = a_{ij} + b_{ij}t|C_{\max}$.

assume that the kth job will start exactly at this time. Then for each permutation one can calculate the total processing time necessary for execution of the second operations of additional jobs from this permutation. Denote the value of this processing time by $L(\pi)$. Let $\pi' = (1, 2, \ldots, n)$. Then $L(\pi') = n$. Consider two permutations, π_{ij} and π_{ji}, differing only in the order of execution of two jobs, J_i and J_j, $i > j$. We will show that $L(\pi_{ij}) > L(\pi_{ji})$. Let these jobs be at positions $k-1$ and k in given permutations. Then we obtain

$$\begin{aligned} L(\pi_{ij}) - L(\pi_{ji}) &= \frac{(E+1)(k-1)}{(E+1)(i-1)} + \frac{(E+1)k}{(E+1)(j-1)} \\ &\quad - \frac{(E+1)(k-1)}{(E+1)(j-1)} - \frac{(E+1)k}{(E+1)(i-1)} \\ &= \frac{i-j}{(i-1)(j-1)} > 0. \end{aligned}$$

Because permutation π' can be obtained from any permutation π by pairwise interchange of two adjacent jobs, which simultaneously decreases the value of $L(\pi)$, the value of $L(\pi)$ will be equal to n if and only if the permutation π is equal to π'. Hence, we obtain a schedule s in which the operations O_{2j}, $j = 1, 2, \ldots, n$, start exactly at times $(j-1)(E+1)$ and their lengths are equal to 1. As a result, we obtain $C_{2j} = (j-1)(E+1) + 1$, $j = 1, 2, \ldots, n$.

There remain only F-jobs to schedule. Denote by $B_1, B_2, \ldots, B_{n-1}$ these sets of jobs that are scheduled in time intervals between C_{1j} and $C_{2,j-1}$, $j = 2, \ldots, n$. Each such interval has the length equal to $C_{2,j-1} - C_{1,j} = E$ and the total processing time of operations from these intervals is equal to $(n-1)E$. Because F-jobs have processing times equal to e_i and $\frac{E}{4} < e_i < \frac{E}{2}$, for each $i \in A$, we conclude that there are only three elements in each such set (otherwise, the sum of processing times of these operations is either lower or greater than E). Thus we have $3(n-1) = 3n-3$ elements in these sets. Denote by B_n the set of jobs processed after operation $O_{2,n}$. As the completion time of the last job from this set is $nE + n$, and $C_{2,n} = (n-1)(E+1) + 1$, this interval also has the length E. In B_n we have $3n - (3n-3) = 3$ elements. But this means that we have defined the partition of the set of all F-jobs into 3-elements subsets $B_1, B_2, \ldots, B_n$ such that $\sum_{i \in B_j} e_i = E$ which implies that the 3-PARTITION problem has a solution. ■

At the end of this section we will prove that if $P \neq NP$, then for the three-machine flow shop problem with simple deteriorating processing times, $F3 \mid p_{ij} = b_{ij}t \mid C_{\max}$, there does not exist a polynomial-time approximation algorithm with the worst case ratio bounded by a constant. (We refer the reader to the book by Blazewicz *et al*,[26] subsection 2.5, for details concerning the worst case analysis of an approximation algorithm.) Moreover, we will show that it is true even in the case when all coefficients b_{ij} of jobs executed on machines M_1 and M_3 are equal to each other. Denote the given problem by $F3|p_{ij} = b_{ij}t, b_{1j} = b_{3j} = b|C_{\max}$. In the proof we use a reduction from the 3-PARTITION problem. (Note that a similar approach was used for the first time by Chen[24] who obtained results similar to ours.)

We will show that the reduction is polynomial with respect to *Length*(Q) and *Max*(Q) for a given instance Q of the 3-PARTITION problem. (We refer the reader to the book by Garey and Johnson,[27] sections 2.1 and 4.2.1, for more details on functions *Length*() and *Max*().)

Let Q be an instance of 3-PARTITION. For any fixed number U, we construct an instance Q_U of the problem $F3|p_{ij} = b_{ij}t, b_{1j} = b_{3j} = b|C_{\max}$ as follows. Let $t_0 = 1$, $|J| = 4n+1$, $b_{1j} = b_{3j} = U^E - 1$ for $1 \leqslant j \leqslant 4n+1$, $b_{2j} = U^{e_j} - 1$ for $1 \leqslant j \leqslant 3n$, $b_{2j} = U^{3E} - 1$ for $3n + 1 \leqslant j \leqslant 4n+1$. For simplicity we will call the first $3n$ jobs *R-jobs* and the remaining jobs *H-jobs*.

Lemma 5 *The instance Q_U can be constructed in time polynomial of Length(Q) and Max(Q).*

Proof: Indeed, $Length(Q) = 3n + \sum_{i \in A} \log_2 e_i$ and $Max(Q) = E$. For the construction of an instance Q_U we have to calculate the values of b_{ij}. For calculating each of them we need $3E-1$ operations of multiplication and one subtraction. Hence for the construction of Q_U we need no more than $O(12n+3)3E$ elementary operations and Q_U can be constructed in time polynomial of *Length*(Q) and *Max*(Q). ■

Note also that $Length(Q_U)$ satisfies inequalities

$$\begin{aligned} Length(Q_U) &\leqslant 3n \log U^E + (n+1) \log U^E \\ &\leqslant 2 \log U \, Max(Q) \, Length(Q) \end{aligned}$$

which imply that $Length(Q_U)$ is polynomial in *Length*(Q) and *Max*(Q).

Lemma 6 *If for an instance Q there is an answer 'Yes', then the optimal makespan for an instance Q_U is less or equal to $U^{(4n+5)E}$.*

Proof: Let there be given a partition of set A into n non-overlapping subsets $A_1, A_2, \ldots, A_n$, such that for any k, $1 \leqslant k \leqslant n$, the equality $\sum_{i \in A_k} e_i = E$ holds.

Let $\pi = (\pi_1, \ldots, i, A_k, j, \ldots, \pi_n)$ mean that the jobs from the set A_k are executed in any order between the ith and the jth jobs. We construct a schedule s for Q_U in the following way: all machines perform jobs without idle times, in the order given by permutation π, $\pi = (3n+1, A_1, 3n+2, A_2, \ldots, 4n, A_n, 4n+1)$, as is shown in Figure 2.

By Lemma 1 we have that the starting time and the completion time of job J_{3n+j}, $1 \leqslant j \leqslant n+1$, on machine M_1 is equal to $S_{1,3n+j} = U^{(4j-4)E}$ and $C_{1,3n+j} = U^{(4j-3)E}$, respectively. From this it follows that on machine M_2 job J_{3n+1} starts at time U^E. Assume that on machine M_2 job

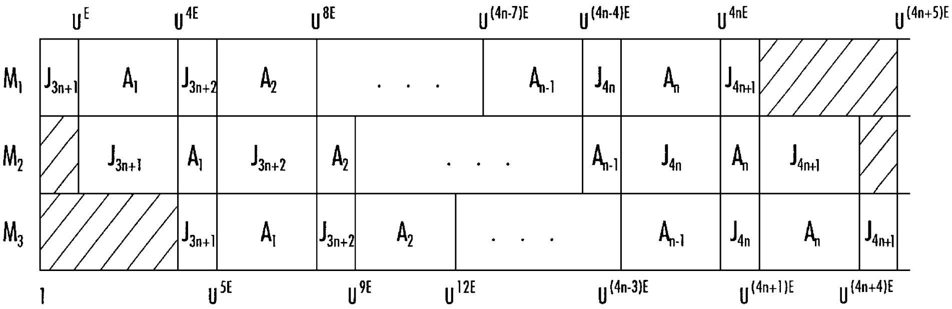

Figure 2 Example schedule for the problem $F3|p_{ij} = b_{ij}t, b_{1j} = b_{3j} = b|C_{\max}$.

J_{3n+j}, $j = 1, 2, \ldots, n+1$, starts at time $U^{(4j\ \ 3)E}$. Then $C_{2,3n+j} = U^{4jE}$. Denote by S_{i,A_j} and C_{i,A_j}, respectively, the starting time and the completion time of all jobs from the set A_j on machine M_i, $i = 1, 2, 3$. Then $S_{1,A_j} = U^{(4j\ \ 3)E}$ and $C_{1,A_j} = U^{4jE}$. Hence, jobs from the set A_j are performed on machine M_2 without idle times, $S_{2,A_j} = U^{4jE}$, $C_{2,A_j} = U^{(4j+1)\Sigma_{i \in A_j} e_i} = U^{(4j+1)E} = U^{(4(j+1)\ \ 3)E}$ and the job J_{3n+j+1} starts on M_2 at the time $U^{(4(j+1)\ \ 3)E}$. By mathematical induction machine M_2 performs all jobs according to permutation π without idle times in time interval $[U^E, U^{(4n+4)E}]$.

In the same way one can prove that machine M_3 finishes the execution of the last job at the time $U^{(4n+5)E}$. ■

Lemma 7 *There exists an optimal solution for an instance Q_U in which the execution of each operation O_{ij} starts at the time $U^{z_{ij}}$, where $z_{ij} \geqslant 0$ is an integer number.*

Proof: Let s be some feasible schedule for the problem Q_U and let O_{ij} be the first operation that starts at the time U^z where z is not integer. From the integrality of all data of Q, from lemma 2 and from the way of constructing of Q_U it follows that if the operation O_{ij} starts at the time being a power of U, then it finishes at the time being a power of U, too. Hence, either the operation O_{ij} is the first on machine M_i, or the preceding operation was finished at the moment of time being a power of U. The same can be said about the preceding operations of job J_j. Thus there exists such a moment U^t of time that $t \leqslant z$ is integer and in the schedule s all operations of job J_j, preceding O_{ij} and operations on machine M_i, preceding O_{ij}, have been finished before time U^t. Assume $S_{ij} = U^t$. The obtained schedule is still feasible and its length will not increase. If we repeat this procedure for the remaining operations which start at moments of time not being powers of U, we obtain a schedule satisfying the conditions given by the lemma. ■

Lemma 8 *If for an instance Q there is an answer 'No', then the optimal makespan for an instance Q_U is $C^*_{\max} \geqslant U^{(4n+5)E+1}$.*

Proof: Because $b_{1i} = U^E\ \ 1$ for any job $J_i \in J$, then machine M_2 starts its work not earlier than at U^E. Then by Lemma 1 we have that

$$C(M_2) \geqslant U^E \prod_{i=1}^{4n+1} (b_{2i} + 1) = U^E U^{nE} U^{3(n+1)E} = U^{4(n+1)E} \tag{2}$$

From (2) and from the fact that $b_{3i} = U^E$, for any job $J_i \in J$, we have that $C^*_{\max} \geqslant U^{(4n+5)E}$. If on machine M_2 there is an idle time after U^E, then $C^*_{\max} > U^{(4n+5)E}$ and by Lemma 7 we obtain $C^*_{\max} \geqslant U^{(4n+5)E+1}$. By Lemma 4 we can assume that in the optimal schedule jobs are performed in the same order on all machines. Let $\pi = (\pi_1, \pi_2, \ldots, \pi_n)$ be a permutation defining the order of jobs in the optimal schedule. We will say that two H-jobs are *adjacent* if, in permutation π, there are no other H-jobs between them. We will consider now a few special cases.

If the first performed job performed is a R-job, then on machine M_2 there is an idle time. Indeed, let it be a job J_j, $j \leqslant 3n$. Then $C_{1,\pi_1} = U^E$, $C_{1,\pi_2} = U^{2E}$, $C_{2,\pi_1} = U^E U^{e_j} < U^{2E}$ and on machine M_2 is an idle time.

If a H-job is performed as the first one in the optimal schedule, then machine M_3 starts its work from the time U^{4E}. Then from (1) we have that

$$C(M_3) \geqslant U^{4E} \prod_{i=1}^{4n+1} (b_{3i} + 1) = U^{4E} U^{(4n+1)E} = U^{(4n+5)E}$$

Thus if on machine M_3 is an idle time after U^{4E}, then by Lemma 7 we obtain $C^*_{\max} \geqslant U^{(4n+5)E+1}$.

If in the optimal schedule no more than two R-jobs are executed between two adjacent H-jobs, then on machine M_3 is an idle time. Indeed, let these two R-jobs be the kth and the $(k+r)$th jobs in permutation π, $r \leqslant 3$ and let the kth job start on machine M_2 at $t \geqslant U^{kE}$. Then $C_{2,\pi_k} = tU^{3E}$, $C_{2\pi_{k+r}} \geqslant tU^{6E+(r\ \ 1)E/4}$. If machine M_3 performs the first $k+r\ \ 1$ jobs without idle times, then $C_{3,\pi_{k+r\ \ 1}} = U^{4E} U^{(k+r\ \ 1)E} = U^{(k+r+3)E}$, $S_{3,\pi_{k+r}} \geqslant C_{2,\pi_{k+r}} \geqslant tU^{6E+(r\ \ 1)E/4} > U^{(k+r+3)E} = C_{3,\pi_{k+r\ \ 1}}$ and there is an idle time on machine M_3 between the $(k+r\ \ 1)$th and the $(k+r)$th jobs.

Because the total number of R-jobs is equal to $n+1$ and the total number of H-jobs is equal to $3n$, in the optimal schedule exactly three R-jobs are executed. Hence, from now on we will assume that in permutation π the H-jobs have indices equal to $4j+1, 0 \leqslant j \leqslant n$.

Let three R-jobs be executed in the optimal schedule between two adjacent H-jobs, $J_{\pi_{4j+1}}$ and $J_{\pi_{4j+5}}$, $0 \leqslant j \leqslant n \quad 1$, and let $\prod_{i=1}^{3}(b_{2,\pi_{4j+1+i}}+1) > U^E$. Then on machine M_3 there is an idle time. Indeed, let operation O_{4j+1} start execution on machine M_2 at $t \geqslant U^{(4j+1)E}$. Then by Lemma 1 we have that $C_{2,\pi_{4j+5}} \geqslant t\prod_{i=0}^{4}(b_{2,\pi_{4j+1+i}}+1) = tU^{6E}\prod_{i=1}^{3}(b_{2,\pi_{j+i}}+1) > tU^{7E}$. If machine M_3 executes the first $4j+4$ jobs without idle times, then $C_{3,\pi_{4j+4}} = U^{4E}U^{(4j+4)E} = U^{(4j+8)E}$, $S_{3,\pi_{4j+5}} \geqslant C_{2,\pi_{4j+5}} > tU^{7E} \geqslant U^{(4j+8)E} = C_{3,\pi_{j+r\ 1}}$ and on machine M_3 between jobs J_{4j+4} and J_{4j+5} is an idle time.

If three R-jobs are executed in the optimal schedule between two adjacent H-jobs, $J_{\pi_{4j+1}}$ and $J_{\pi_{4j+5}}$, $0 \leqslant j \leqslant n \quad 1, \prod_{i=1}^{3}(b_{2,\pi_{4j+1+i}}+1) < U^E$ and if there are no idle times on machine M_3, then on machine M_2 there is an idle time.

Indeed, let the operation O_{4j+1} start on machine M_1 at $t \geqslant U^{4jE}$. Then $C_{1,\pi_{4j+5}} = tU^{5E}$. Because on machine M_3 there are no idle times, then $\prod_{i=1}^{3}(b_{2,\pi_{4j+1+i}}+1) \leqslant U^E$ for all $0 \leqslant j \leqslant n$. Hence, $C_{2,\pi_{4j}} \leqslant U^E\prod_{k=1}^{j}U^{3E}\prod_{i=1}^{3}(b_{2,\pi_{4k+1+i}}+1) \leqslant U^{(4j+1)E} \leqslant tU^E$ and $C_{2,\pi_{4j+4}} = tU^EU^{3E}\prod_{i=1}^{3}(b_{2,\pi_{4j+1+i}}+1) < tU^{5E}$. We obtain $S_{1,\pi_{4j+5}} \geqslant C_{1,\pi_{4j+5}} > C_{2,\pi_{4j+4}}$ and on machine M_2 is an idle time.

Let three R-jobs be executed in the optimal schedule between two adjacent H-jobs, $J_{\pi_{4j+1}}$ and $j_{\pi_{4j+5}}$, $0 \leqslant j \leqslant n \quad 1$, and let $\prod_{i=1}^{3}(b_{2,\pi_{4j+1+i}}+1) < U^E$. Then for the instance Q we can assume $A_j = \{\pi_{4j+2}, \pi_{4j+3}, \pi_{4j+4}\}, 0 \leqslant j \leqslant n \quad 1$. It is easy to see that partition of the set A into n such subsets A_j is the wanted partition in the 3-PARTITION problem. We have obtained a contradiction with the assumption that for Q there is an answer 'No'. ■

Theorem 2 *If $P \neq NP$, then for the problem $F3 \mid p_{ij} = b_{ij}t, b_{1j} = b_{3j} = b \mid C_{\max}$ there does not exist a polynomial-time approximation algorithm with the worst case ratio bounded by a constant.*

Proof: Suppose that there exists a polynomial-time approximation algorithm A with ratio $R_A < U$, U is some fixed number. Then using an instance Q of the problem 3-PARTITION we construct an instance Q_U of the problem $F3 \mid p_{ij} = b_{ij}t, b_{1j} = b_{3j} = b \mid C_{\max}$ and solve it by algorithm A. From Lemma 6 it follows that, if for Q there is an answer 'Yes', then makespan obtained by algorithm A for Q_U is $C^*_{\max} < U^{(4n+5)E}U = U^{(4n+5)E+1}$. From Lemma 8 it follows that, if for Q there is an answer 'No', then makespan obtained by algorithm A for Q_U is $C^*_{\max} \geqslant U^{(4n+5)E+1}$. By Lemma 5 we have that the time of execution of algorithm A is bounded by a polynomial of $Length(Q)$ and $Max(Q)$. But this means that we obtain a pseudo-polynomial algorithm for solving the strongly NP-complete 3-PARTITION problem which is contradictory to the hypothesis $P \neq NP$. ■

Open shop problem

In this section we will establish the time complexity of two cases of the open shop problem: with simple linear processing times and with proportional processing times.

We start with the definition of two ordinarily NP-complete problems, PARTITION and SUBSET PRODUCT (Chen,[24] Garey and Johnson,[27] pp 223–224).

PARTITION. Given a set $A = \{1, 2, \ldots, n\}$, an integer E, and positive integers e_i such that $\sum_{i \in A} e_i = 2E$, does there exist a subset $A' \subseteq A$ such that $\sum_{i \in A'} e_i = \sum_{i \in A \setminus A'} e_i = E$?

SUBSET PRODUCT. Given a set $A = \{1, 2, \ldots, n\}$, an integer E, and positive integers $e_i, i \in A$, does there exist a subset $A' \subseteq A$ such that $\prod_{i \in A'} e_i = E$?

We consider first the open shop scheduling problem with two machines and linear deterioration.

Theorem 3 *The problem $O2 \mid p_{ij} = a_{ij} + b_{ij}t \mid C_{\max}$ is NP-hard in the ordinary sense.*

Proof: We will show that the PARTITION problem is reducible to the problem $O2 \mid p_{ij} = a_{ij} + b_{ij}t \mid C_{\max}$. The reduction is as follows. Let $t_0 = 1, |J| = n+1$, $p_{1i} = p_{2i} = e_i$ for $1 \leqslant i \leqslant n$ and $p_{1,n+1} = p_{2,n+1} = Et$. We will show that the PARTITION problem has a solution if and only if the makespan for the constructed instance is not greater than $(E+1)^2 + E$.

($\Rightarrow$) Assume that PARTITION problem has a solution, ie there exists $A' \subseteq A$ such that $\sum_{i \in A'} e_i = E$. Then we schedule jobs in the following way: on machine M_1 job J_{n+1} is executed as the first one and then followed by the remaining jobs in an arbitrary order, and on machine M_2 jobs with indices from A' are executed first (in an arbitrary order), then comes job J_{n+1}, followed by the remaining jobs in an arbitrary order (see Figure 3). It is easy to see that the length of the schedule is equal to $(E+1)^2 + E$.

($\Leftarrow$) Assume now that there exists a schedule with the makespan not greater than $(E+1)^2 + E$. Without loss of generality, we can assume that all jobs start and finish at integer moments of time. Note that there exists one operation of job J_{n+1} such that it starts not earlier than $E+1$. Let this operation be executed on machine M_2 and let the operation start at time $E+1+r, r \geqslant 0$. Then $C_{2,n+1} =$

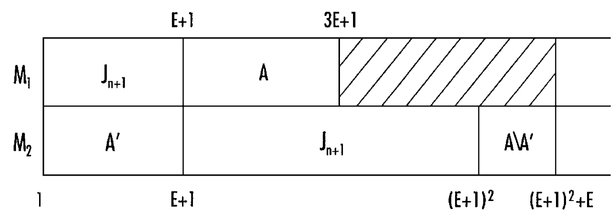

Figure 3 Example schedule for the problem $O2|p_{ij} = a_{ij} + b_{ij}t|C_{\max}$.

$E+1+r+E(E+1+r)=(E+1+r)(E+1)=(E+1)^2+rE+r$. Because up to the time $E+1+r$ the machine M_2 can work for no more that $E+r$ units of time, then the total length (after the completion of job J_{n+1}) of unexecuted operations is not lower than $E \quad r$. Thus, $C(M_2) \geqslant C_{2,n+1}+E \quad r=(E+1)^2+(r+1)E$. Because we have assumed that $C_{\max}(s) \leqslant (E+1)^2+E$, we obtain $r=0$ and machine M_2 starts job J_{n+1} exactly at time $E+1$. To finish the proof, it is sufficient to note that the completion time of the last operation on machine M_2 is equal to $(E+1)^2+E$ only if all jobs are executed starting from time 1 up to $(E+1)^2+E$ without idle times. Denote by A' the set of indices of jobs that are executed on machine M_2 before the job J_{n+1}. Since $\sum_{i \in A'} e_i = E$, the PARTITION problem has a solution. ■

Now we will prove that the three-machine open shop problem with simple linear deterioration is computationally intractable, even if jobs have got equal deterioration rates on the third machine.

Theorem 4 *The problem $O3 \mid p_{ij}=b_{ij}t, b_{3j}=b \mid C_{\max}$ is NP-hard in the ordinary sense.*

Proof: We construct a reduction from the SUBSET PRODUCT problem to the problem $O3 \mid p_{ij}=b_{ij}t, b_{3j}=b \mid C_{\max}$. Denote $\prod_{i \in A} e_i = E_0$, $\bar{E}=2E_0$. Let $t_0=1$, $\mid J \mid = n+4$,

$$
\begin{array}{lll}
b_{1i}=0 & b_{2i}=e_i^2 \quad 1 & b_{3i}=\bar{E}^2 \quad 1 \\
 & & 1 \leqslant i \leqslant n \\
b_{1,n+1}=0 & b_{2,n+1}=\bar{E}^2/E^2 \quad 1 & b_{3,n+1}=\bar{E}^2 \quad 1 \\
b_{1,n+2}=0 & b_{2,n+2}=4\bar{E}^2 \quad 1 & b_{3,n+2}=\bar{E}^2 \quad 1 \\
b_{1,n+3}=\bar{E}^{n+3} \quad 1 & b_{2,n+3}=\bar{E}^{n+3} \quad 1 & b_{3,n+3}=\bar{E}^2 \quad 1 \\
b_{1,n+4}=\bar{E}^{n+5} \quad 1 & b_{2,n+4}=\bar{E}^{n+1} \quad 1 & b_{3,n+4}=\bar{E}^2 \quad 1
\end{array}
$$

Without loss of generality we can assume that n is an even number.

We will show that for the constructed instance there exists a schedule s for which $C(s) \leqslant \bar{E}^{2n+8}$, if and only if in the set A there exists a subset $A' \subseteq A$ such that the product of its elements is equal to E.

($\Rightarrow$) Let there exist such a subset A'. Then schedule s can be constructed in the following way. Machine M_1 executes first job J_{n+3}, and then job J_{n+4}. On machine M_2 jobs with indices from A' are executed in an arbitrary order, followed by jobs J_{n+1}, J_{n+4}, J_{n+3} and J_{n+2}, and then by jobs from the set $A \backslash A'$, again in an arbitrary order. Machine M_3 performs jobs in the following order: J_{n+4}, $J_1, J_2, \ldots, J_n, J_{n+1}, J_{n+2}, J_{n+3}$ (see in Figure 4).

It is easy to check that all machines work without idle times in the time interval $[1, \bar{E}^{2n+8}]$ and that $C_{\max}(s) \leqslant \bar{E}^{2n+8}$.

($\Leftarrow$) Let there exist a schedule s such that $C(s) \leqslant \bar{E}^{2n+8}$. Suppose that all machines start to work from time 1. Let us calculate completion times of each machine. From (1) we have that

$$C(M_1)=(b_{1,n+3}+1)(b_{1,n+4}+1)=\bar{E}^{n+3}\bar{E}^{n+5}=\bar{E}^{2n+8}$$

$$C(M_2)=\prod_{i=1}^{n+4}(b_{2i}+1)=\frac{\bar{E}^2}{E^2}\cdot \bar{E}^{n+1}\cdot \bar{E}^{n+3}\cdot 4E^2\prod_{i=1}^{n} e_i^2=\bar{E}^{2n+8}$$

$$C(M_3)=\prod_{i=1}^{n+4}(b_{3i+1})=\bar{E}^{2n+8}$$

From the above it follows that in schedule s all machines work without idle times in the time interval $[1, \bar{E}^{2n+8}]$. Note that from this fact, and from $b_{3i}=\bar{E}^2 \quad 1$, $1 \leqslant i \leqslant n+4$, it follows that on machine M_3 all jobs start and finish at the moments of time being even powers of $\bar{E}$. Let us calculate completion times of jobs J_{n+4} and J_{n+3} assuming that they start at time 1 and are executed without idle times. We have

$$C_{n+4}=\prod_{i=1}^{3}(b_{i,n+4}+1)=\bar{E}^{n+5}\cdot \bar{E}^{n+1}\cdot \bar{E}^2=\bar{E}^{2n+8}$$
$$C_{n+3}=\prod_{i=1}^{3}(b_{i,n+3}+1)=\bar{E}^{n+3}\cdot \bar{E}^{n+3}\cdot \bar{E}^2=\bar{E}^{2n+8}$$

From the above it is clear that in schedule s jobs J_{n+4} and J_{n+3} are executed without idle times, starting from time 1. We will show that then jobs J_{n+4} and J_{n+3} are executed on machine M_2 consecutively in the interval $[\bar{E}^2, \bar{E}^{2n+6}]$.

Indeed, if the first from these jobs, J_{n+4}, starts on machine M_2 at time $0<t<\bar{E}^2$, then this job cannot be executed on machine M_3 in the interval $[1, \bar{E}^2]$. Analogously, if the job J_{n+3} is completed on machine M_2 at $\bar{E}^{2n+6}<t<\bar{E}^{2n+8}$, then this job, performed on machine M_3, cannot be executed in the interval $[\bar{E}^{2n+6}, \bar{E}^{2n+8}]$. Let

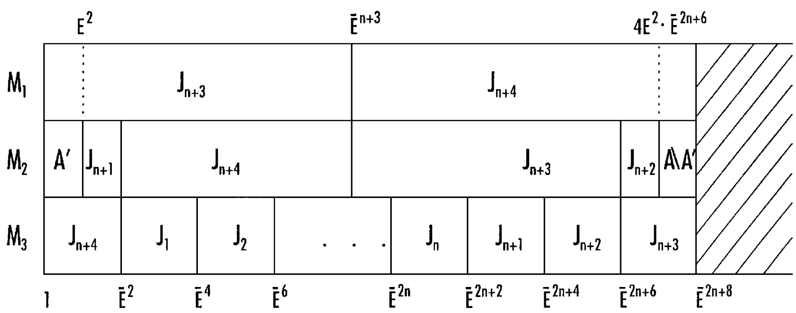

Figure 4 Example schedule for the problem $O3|p_{ij}=b_{ij}t, b_{3j}=b|C_{\max}$.

job J_{n+3} start on machine M_2 at time 1. Then job J_{n+4} is executed as the first on machine M_1. Hence $S_{1,n+3} = \overline{E}^{n+5}$, $C_{2,n+3} = \overline{E}^{n+3}$, and the job J_{n+3} starts on machine M_3 at time $\overline{E}^{n+3}$. We obtain a contradiction with the fact that in schedule s all jobs performed on machine M_3 start and finish at moments of time being even powers of $\overline{E}$. Assume now that job J_{n+4} starts on machine M_2 at time 1. Then job J_{n+3} is executed as the first on machine M_1. Hence $S_{1,n+4} = \overline{E}^{n+3}$, $C_{2,n+4} = \overline{E}^{n+1}$, and the job J_{n+4} starts on machine M_3 at time $\overline{E}^{n+1}$ which is not an even power of $\overline{E}$. Hence in schedule s neither job J_{n+3} nor job J_{n+4} starts on machine M_2 at time 1 or finishes at time $\overline{E}^2$.

Hence, $n+2$ remaining jobs on machine M_2 are executed in time intervals $[1, \overline{E}^2]$ and $[\overline{E}^{2n+6}, \overline{E}^{2n+8}]$. Because

$$(b_{2,n+1} + 1)(b_{2,n+2} + 1) = \frac{\overline{E}^2}{\underline{E}^2}.4E^2 = 4\overline{E}^2 > \overline{E}^2,$$

then jobs J_{n+1} and J_{n+2} cannot be executed in schedule s in one interval. Let the job J_{n+1} be executed in schedule s in the interval $[1, \overline{E}^2]$. Denote by A_1 the set of indices of all jobs executed in this interval. Assume that $A' = A_1 \backslash \{n+1\}$. Then we have $\prod_{i \in A_1}(b_i + 1) = \bar{E}^2 = 4\bar{E}_0^2$. On the other hand, from (1) it follows that $\bar{E}^2 = 4E_0^2 = 4\frac{E_0^2}{E^2}\prod_{i \in A'} e_i^2$. Thus $\prod_{i \in A'} e_i = E$, and the wanted set A' has been constructed which implies that the SUBSET PRODUCT problem has a solution. ■

Conclusions

We have considered the flow shop and open shop scheduling problems on two or more machines with simple linear, linear or proportional deterioration subject to minimizing the maximum completion time (makespan). We have proved that linear deterioration in both cases, flow shop and open shop, is computationally difficult: the two-machine flow shop problem with linear deterioration is strongly NP-hard, while the two-machine open shop problem with linear deterioration is ordinarily NP-hard. We have shown that for the three machine flow shop problem there does not exist a polynomial-time approximation algorithm with the worst case ratio bounded by a constant, unless $P = NP$. We have proved that the three-machine open shop problem with simple linear deterioration is ordinarily NP-hard, even if the jobs have got equal deterioration rates on one of the machines available. We have also shown that optimal schedules for a two- or three-machine flow shop with proportional processing times of jobs are permutation schedules.

One can point out two main directions in the research of the scheduling of jobs with time-dependent processing times on dedicated machines. The first one is to consider other, not yet solved, problems in order to discover their complexity status. One of the important candidates is the job shop problem. We conjecture that this problem for already two machines and simple linear deterioration is at least NP-hard in the ordinary sense. The second area of research is the construction and investigation of good polynomial approximation algorithms or approximation schemata for time-dependent scheduling problems. Some results are already known. For example, the worst case results for SDR (Smallest Deterioration Rate) and LDR (Largest Deterioration Rate) heuristics of scheduling jobs with bounded simple linear deterioration on parallel identical machines (Hsieh and Bricker,[28] Mosheiov[25]) or approximation schemata for bounded non-linear deterioration and a single machine (Cai *et al*,[29] Kovalyov and Kubiak[30]). We conjecture, however, that if we deal with unbounded deterioration, then there does not exist an approximation algorithm with the worst case ratio bounded by a constant, unless $P = NP$.

Acknowledgements—The research of Dr Kononov has been supported by Russian Fund of Fundamental Research grant no. 99-01-00510. The research of Dr Gawiejnowicz has been partially supported by KBN (Polish Scientific Committee) grant no. 8T11A 01618.

The authors wish to thank to two anonymous referees for valuable comments on an earlier version of the paper.

References

1 Conway RW, Maxwell WL and Miller LW (1967). *Theory of Scheduling*. Addison-Wesley: Reading, MA.
2 Sriskandarajah C and Goyal SK (1989). Scheduling of a two-machine flow-shop with processing time linearly dependent on job waiting-time. *J Opl Res Soc* **40**: 907–921.
3 Sriskandarajah C and Wagneur E (1991). The hierarchical control of the two-machine flowshop with state dependent processing times. Complexity analysis and approximate algorithms. *Infor* **29**: 193–205.
4 Wagneur E and Sriskandarajah C (1993). Optimal control of a class of DEDS: flow-shops with state-dependent processing times. *Discr Event Dyn Syst. Theor Appl* **3**: 397–425.
5 Wagneur E and Sriskandarajah C (1993). The two-machine permutation flow shop with state-dependent processing times. *Nav Res Logist* **40**: 697–717.
6 Finke G and Jiang H (1997). A variant of the permutation flow shop model with variable processing times. *Discr Appl Math* **76**: 123–140.
7 Melnikov OI and Shafransky YM (1979). Parametric problem of scheduling theory. *Kibernetika* **6**: 53–57. (Engl. transl. 1980 *Cybernetics* **15**: 352–357.)
8 Alidaee B and Womer NK (1999). Scheduling with time dependent processing times: review and extensions. *J Opl Res Soc* **50**: 711–720.
9 Gawiejnowicz S (1996). Brief survey of continuous models of scheduling. *Found Comput Dec Sci* **21**: 81–100.
10 Browne S and Yechiali U (1990). Scheduling deteriorating jobs on a single processor. *Opns Res* **38**: 495–498.
11 Gawiejnowicz S and Pankowska L (1995). Scheduling jobs with varying processing times. *Inform Process Lett* **54**: 175–178.
12 Kononov A (1996). Combinatorial complexity of scheduling jobs with simple linear deterioration. *Discr Anal Oper Res* **3**: 15–32 (in Russian).

13 Kononov A (1997). Scheduling problems with linear processing times. In: Zimmermann U *et al* (eds). *Operations Research Proceedings 1996. Selected Papers of the Symposium on Operations Research (SOR 96).* Springer: Berlin, pp 208–212.
14 Mosheiov G (1991). V-shaped policies to schedule deteriorating jobs. *Opns Res* **39**: 979–991.
15 Mosheiov G (1994). Scheduling jobs under simple linear deterioration. *Comput Opns Res* **21**: 653–659.
16 Alidaee B (1990). A heuristic solution procedure to minimize makespan on a single machine with non-linear cost. *J Opl Res Soc* **41**: 1065–1068.
17 Gupta JND and Gupta SK (1988). Single facility scheduling with nonlinear processing times. *Compu Ind Engng* **14**: 387–393.
18 Kononov A (1997). On schedules of a single machine jobs with processing times nonlinear in time. In: Korshunov AD (ed). *Operations Research and Discrete Analysis*. Kluwer: Dordrecht, pp 109–122.
19 Kononov A (1998). Single machine scheduling problems with processing times proportional to an arbitrary function. *Discr Anal Oper Res* **5**: 17–37 (in Russian).
20 Kubiak W and van de Velde S (1998). Scheduling deteriorating jobs to minimize makespan. *Nav Res Logist* **45**: 511–523.
21 Kunnathur AS and Gupta SK (1990). Minimizing the makespan with late start penalties added to processing times in a single facility scheduling problem. *Euro J Opl Res* **47**: 56–64.
22 Sundararaghavan PS and Kunnathur AS (1994). Single machine scheduling with start time dependent processing times: some solvable cases. *Euro J Opl Res* **78**: 394–403.
23 Cheng TCE and Ding Q (1998). The complexity of scheduling starting time dependent tasks with release times. *Inform Process Lett* **65**: 75–79.
24 Chen Z-L (1996). Parallel machine scheduling with time dependent processing times. *Discr Appl Math* **70**: 81–93 (erratum: *Discr Appl Math* **75**: 103).
25 Mosheiov G (1998). Scheduling linearly deteriorating jobs on multiple machines. *Infor* **36**: 205–214.
26 Blażewicz J *et al* (1996). *Scheduling Computer and Manufacturing Processes.* Springer: Berlin.
27 Garey MR and Johnson DS (1979). *Computers and Intractability: A Guide to the Theory of NP-Completeness.* WH Freeman: San Francisco.
28 Hsieh Y-C and Bricker DL (1997). Scheduling deteriorating jobs on multiple machines. *Computer Ind Engng* **32**: 727–734.
29 Cai J-Y, Cai P and Zhu Y (1998). On a scheduling problem of time deteriorating jobs. *J Complex* **14**: 190–209.
30 Kovalyov MY and Kubiak W (1998). A fully polynomial approximation scheme for minimizing makespan of deteriorating jobs. *J Heurist* **3**: 287–297.

Received December 1999;
accepted December 2000 after two revisions